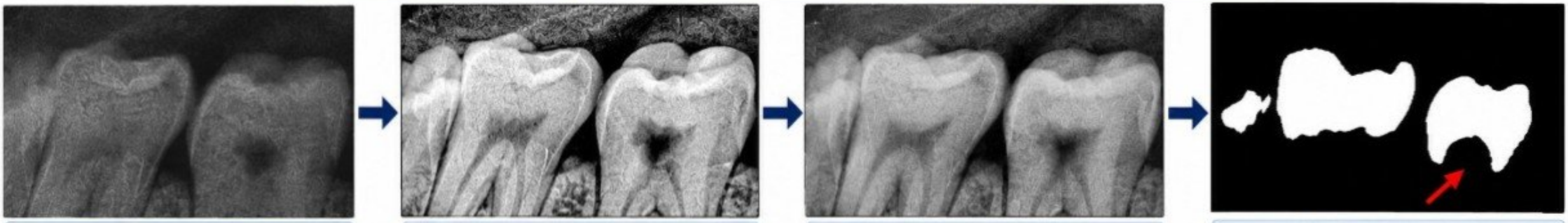
1. Dental X-Ray Enhancement and Cavity Region Extraction



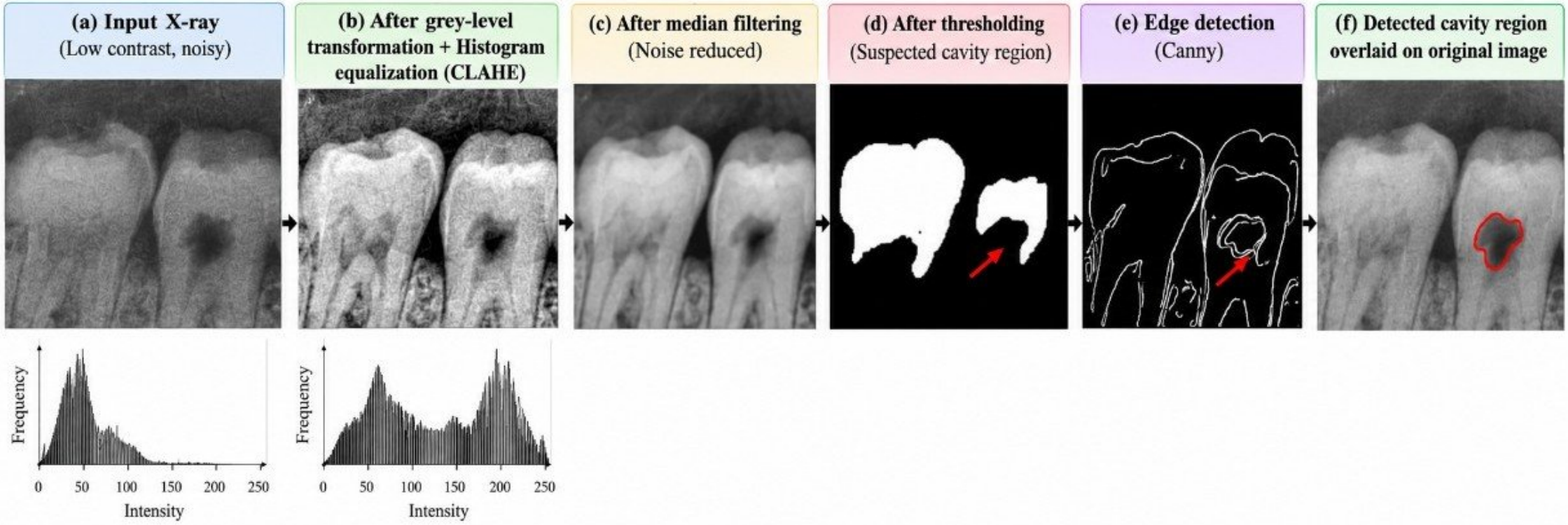
(a) Input X-ray
(Low contrast, noisy)

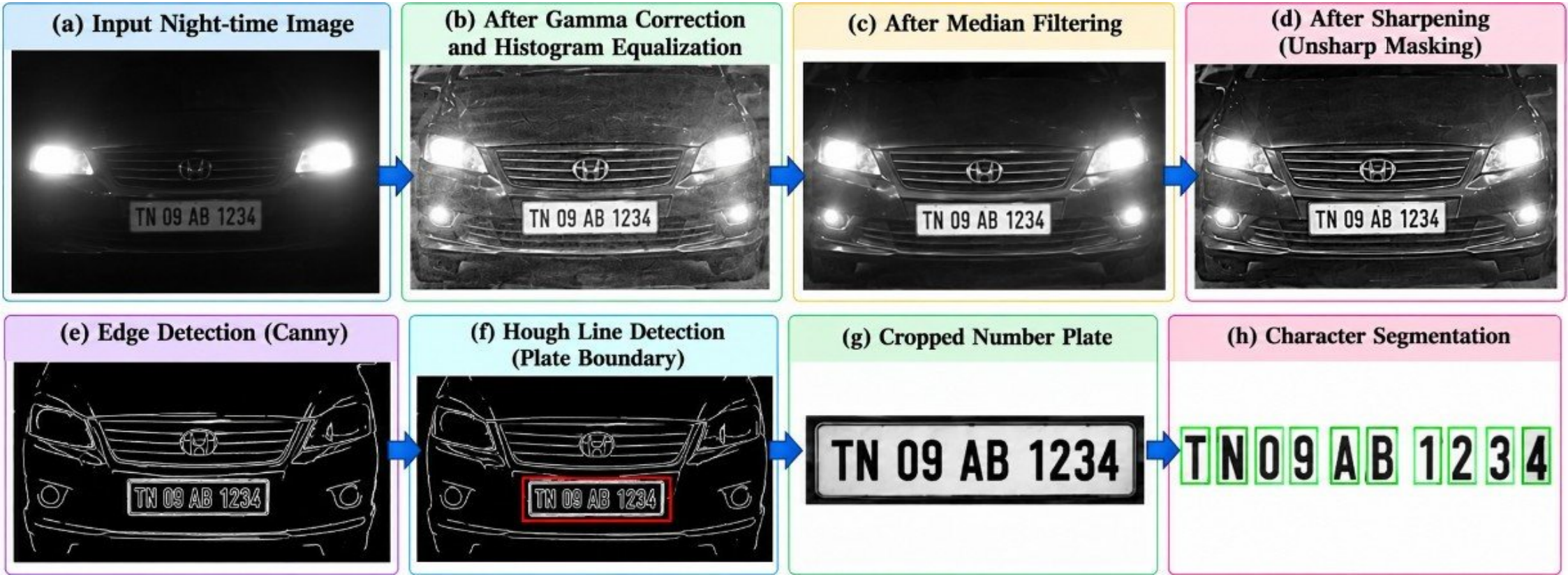
(b) After grey-level transformation + Histogram equalization (CLAHE)

(c) After median filtering
(Noise reduced)

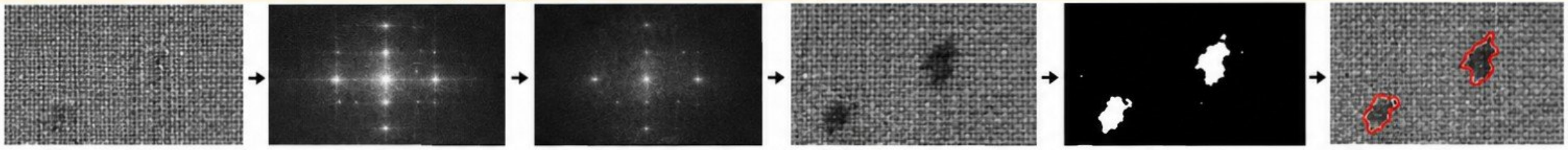
(d) After thresholding
(Suspected cavity region)

1. Dental X-Ray Enhancement and Cavity Region Extraction




(a) Input Night-time Image
(b) After Gamma Correction and Histogram Equalization
(c) After Median Filtering
(d) After Sharpening (Unsharp Masking)
(e) Edge Detection (Canny)
(f) Hough Line Detection (Plate Boundary)
(g) Cropped Number Plate
(h) Character Segmentation
TN 09 AB 1234
TN09AB1234

3. Textile Fabric Defect Detection Using Frequency-Domain Processing



(a) Input fabric image (Regular weave, periodic noise)

(b) Fourier spectrum (with periodic peaks)

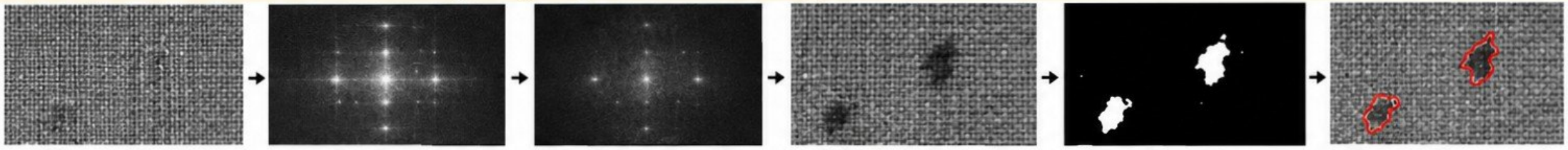
(c) After notch filtering (Periodic noise removed)

(d) Reconstructed image after filtering

(e) Thresholding (Defect regions)

(f) Detected defects outlined on original image

3. Textile Fabric Defect Detection Using Frequency-Domain Processing



(a) Input fabric image (Regular weave, periodic noise)

(b) Fourier spectrum (with periodic peaks)

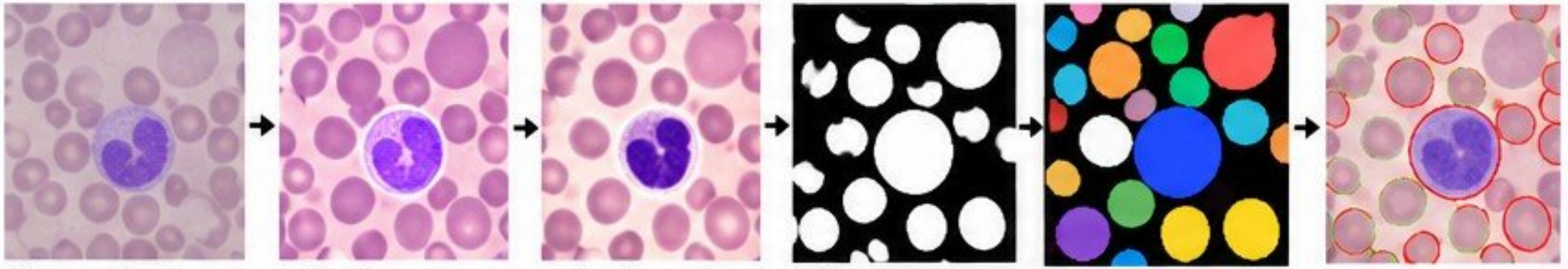
(c) After notch filtering (Periodic noise removed)

(d) Reconstructed image after filtering

(e) Thresholding (Defect regions)

(f) Detected defects outlined on original image

4. Blood Cell Segmentation in Microscopic Images



(a) Input blood smear (Uneven staining, low contrast)

(b) After contrast enhancement (CLAHE)

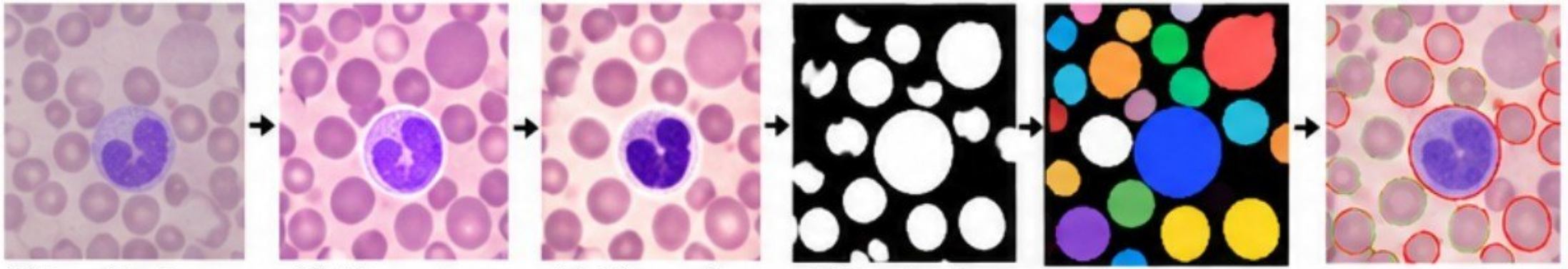
(c) After median filtering

(d) Thresholding (Binary image)

(e) Watershed segmentation (Separated touching cells)

(f) Detected cell boundaries overlaid on original image

4. Blood Cell Segmentation in Microscopic Images



(a) Input blood smear (Uneven staining, low contrast)

(b) After contrast enhancement (CLAHE)

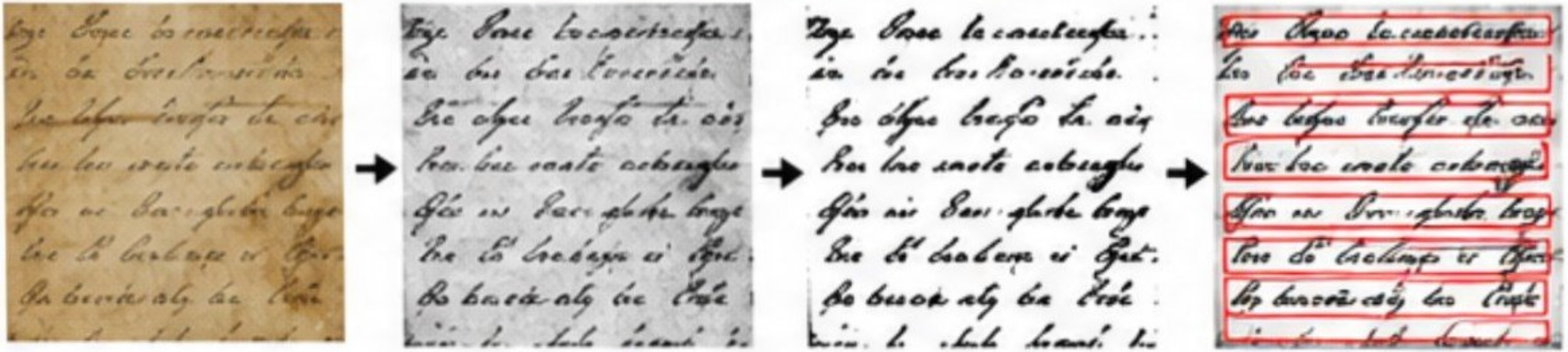
(c) After median filtering

(d) Thresholding (Binary image)

(e) Watershed segmentation (Separated touching cells)

(f) Detected cell boundaries overlaid on original image

5. Text Extraction from Degraded Historical Documents



(a) Input document (Yellowed, faded, stains)

(b) After enhancement (Contrast improved)

(c) Adaptive thresholding (Binary)

(d) Extracted text regions (Contours)